STUDENT RECORD SYSTEM

Requirements Traceability and Backlog Review Report

Team 11

Executive Summary

This report reviews the Student Record System Software Requirements Specification (SRS) against the project's GitHub backlog and Requirements Traceability Matrix (RTM). The review covers 25 functional requirements (F-001–F-025), 7 non-functional requirements (NF-001–NF-007), and 5 security requirements (SR-001–SR-005).

The existing backlog contains issues for student-record management, academic records, attendance, reports, authentication, access control, persistence, validation, and security/audit logging. Additional issues were created for required student fields and duplicate-ID rejection (#15), accessibility (#16), performance and response-time testing (#17), and secure logout/session termination (#18). Issue #14 is the general testing backlog item and should be checked to ensure its scope includes verification of the full set of functional and non-functional requirements.

The RTM lists test-case identifiers for the requirements, but its current test-execution status is N (Not Tested) for all listed rows. Therefore, the existence of a test-case ID or GitHub issue is not evidence that a requirement has been implemented or has passed testing. Some mappings—especially F-025 account management, F-023 data consistency, and NF-002 reliability—need further clarification or verification.

1. Introduction

The Student Record System is intended to support the management and retrieval of student and academic information. This report documents a requirements traceability review: it connects SRS requirements to GitHub backlog items and the test cases listed in the RTM, identifies areas requiring clarification, and states the limits of the available verification evidence.

1.1 Objectives

- Map each SRS requirement to one or more relevant GitHub issues and RTM test-case identifiers.
- Record backlog items added to make specific requirements more explicit.
- Distinguish requirement/backlog coverage from implementation and test-execution status.
- Identify gaps that need team confirmation or further verification before final sign-off.

1.2 Scope and limitations

The review is based on the known SRS requirement IDs, the GitHub issue titles and descriptions discussed during the review, and the RTM entries supplied for this work. It is a documentation and traceability review, not an independent code audit. Where implementation evidence, execution logs, screenshots, or measured results were not provided, the report does not infer that a requirement is complete.

2. Review Method

1. Identify the requirement IDs in the SRS: F-001–F-025, NF-001–NF-007, and SR-001–SR-005.
2. Compare the requirements with the existing GitHub issues #1–#13 and the testing issue #14.
3. Record additional issues #15–#18 and clarification comments raised for Issues #6 and #10.
4. Map requirements to the RTM test-case IDs and preserve the test-execution status separately from backlog coverage.
5. Flag incomplete or uncertain mappings rather than treating them as verified.

3. GitHub Backlog Overview

Issue	Title	Traceability note
#1	Implement student record creation	Student record creation; linked to required fields and duplicate-ID rejection through #15.
#2	Implement student record viewing and search	Authorized viewing and search by student ID/name.
#3	Implement student record updating	Authorized student record updates.
#4	Implement student record deletion	Deletion flow; SRS requirement mapping needs clarification because no explicit deletion requirement appears in the listed functional requirements.
#5	Implement user authentication and login	Authentication, password protection, session timeout and failed-login controls.
#6	Implement role-based access control (RBAC)	Role permissions and access restrictions; F-025 account create/modify/deactivate requirements need explicit tracking.
#7	Implement subject and grade management	Subject maintenance, subject assignment, grade entry, grade validation and grade updates.
#8	Implement GPA and CGPA calculation	GPA/CGPA calculation based on grades.
#9	Implement attendance tracking	Attendance recording, updating and percentage calculation.
#10	Implement data persistence and storage	Persistent storage and data integrity; verify consistency across related student, subject, grade and attendance records.
#11	Implement reports and record summaries	Academic reports, grade sheets and attendance summaries.
#12	Implement input validation and error handling	Required-field validation, invalid-input handling, grade validation and valid record references.
#13	Implement security controls and audit logging	Data protection, audit logging, password security and failed-login safeguards.
#14	Test functional and non-functional requirements	Cross-cutting testing issue covering all applicable functional, non-functional and security requirements.
#15	Include required student fields and duplicate-ID rejection	Additional issue mapped to F-001, F-004 and F-022.
#16	Implement accessibility support	Additional issue mapped to NF-005.

#17	Implement performance requirements and response-time testing	Additional issue mapped to NF-001 and NF-006.
#18	Implement secure logout and session termination	Additional issue mapped to F-024; session timeout is separately specified by SR-003.

4. Requirements Traceability Matrix

The table below consolidates the known requirement-to-issue mappings and the test-case IDs listed in the RTM. Coverage is assessed separately from test execution. “Partial / clarify” means that the issue is related to the requirement but the available description, acceptance criteria, or verification evidence needs further confirmation.

Test status legend: N = Not Tested; P = Passed; A = Action Required. The supplied RTM currently marks all rows N. Do not change N to P unless a test has actually been executed and the result recorded.

Req. ID	Requirement summary	GitHub issue(s)	RTM test case	Coverage / verification note	Test status
F-001	Create student record with required information	#1, #15	TC-STD-01	Verify required fields and successful record creation.	N
F-002	Authorized users can view student records	#2	TC-STD-02	Verify authorized viewing.	N
F-003	Administrators can update student records	#3	TC-STD-03	Verify authorized updates.	N
F-004	Reject duplicate student IDs	#1, #12, #15	TC-STD-04	Verify duplicate-ID rejection.	N
F-005	Administrators maintain subjects	#7	TC-SUB-01	Verify subject creation and maintenance.	N
F-006	Users can view subjects as permitted	#7	TC-SUB-02	Verify subject viewing permissions.	N
F-007	Assign subjects to students	#7	TC-SUB-03	Verify subject assignment.	N
F-008	Faculty enter grades	#7	TC-GRD-01	Verify authorized grade entry.	N
F-009	Validate grade values	#7, #12	TC-GRD-02	Verify grade validation boundaries.	N
F-010	Update grades with appropriate authorization	#7	TC-GRD-03	Verify authorized grade updates.	N
F-011	Calculate GPA and CGPA	#8	TC-GRD-04	Verify calculation formulas and edge cases.	N
F-012	Faculty record attendance	#9	TC-ATT-01	Verify attendance recording.	N

F-013	Update attendance records	#9	TC-ATT-02	Verify attendance updates.	N
F-014	Calculate attendance percentage	#9	TC-ATT-03	Verify attendance percentage calculations.	N
F-015	Search by student ID or name	#2	TC-SRCH-01	Verify search by ID and name.	N
F-016	Generate academic reports containing grades and attendance	#11	TC-RPT-01	Verify report contents and generation.	N
F-017	Generate grade sheets and attendance summaries	#11	TC-RPT-02	Verify grade sheets and attendance summaries.	N
F-018	Authenticate users	#5	TC-AUTH-01	Verify authentication before access.	N
F-019	Enforce role-based permissions	#6	TC-AUTH-02	Verify permissions for each defined role.	N
F-020	Prevent students from modifying protected academic records	#6	TC-AUTH-03	Verify unauthorized modifications are blocked.	N
F-021	Persist records in storage	#10	TC-DATA-01	Verify persistence, including after application restart.	N
F-022	Validate required fields	#1, #12, #15	TC-DATA-02	Verify required-field validation before storage.	N
F-023	Maintain consistency across students, subjects, grades and attendance	#10, #12	TC-DATA-03	Verify invalid references are rejected; clarify and test update/delete consistency rules.	N
F-024	Logout terminates the user session	#18	TC-EXIT-01	Verify that protected pages cannot be accessed after logout.	N
F-025	Administrators create, modify and deactivate authorized user accounts	#6 — clarification required	TC-ADMIN-01	Account lifecycle management is distinct from RBAC; verify implementation and test coverage.	N
NF-001	Dashboard/list pages meet the response-time target under normal load	#17	TC-Perf-01	Verify the 95th-percentile response time is ≤ 3 seconds under normal load.	N
NF-002	Reliable operation without unexpected termination or data loss	#10	TC-REL-01	Define and test reliability, failure and recovery scenarios.	N
NF-003	Protect personal information and sensitive data	#13, #5	TC-Sec-01	Verify data protection and the absence of plaintext sensitive information where prohibited.	N
NF-004	Audit relevant actions and retain logs for the required period	#13	TC-OPS-01	Verify audit events, actor, action, timestamp and five-year retention.	N
NF-	Accessibility support for keyboard,	#16	TC-UX-01	Verify applicable WCAG 2.1 AA accessibility	N

005	screen readers and contrast			criteria.	
NF-006	Support expected record volume without significant degradation	#17, #10	TC-Perf-02	Define the expected dataset and verify performance under that load.	N
NF-007	Enforce distinct permissions for each role	#6	TC-Sec-02	Verify the role-permission matrix.	N
SR-001	Protect information from unauthorized access	#13, #6	TC-Sec-01	Verify access controls and safeguards for student information.	N
SR-002	Store passwords using a suitable secure hash	#5, #13	TC-Sec-02	Verify password hashing; passwords must not be stored in plaintext.	N
SR-003	Apply the specified session timeout	#5	TC-Sec-03	Verify session termination after 15 minutes of inactivity.	N
SR-004	Handle repeated failed login attempts and apply restriction after five failures	#5, #13	TC-Sec-04	Verify failed-login logging and restriction after five consecutive failures.	N
SR-005	Restrict access according to user role	#6	TC-Sec-05	Verify access permissions for Admin, Faculty, Authorized Staff and Student.	N

5. Key Findings and Actions

5.1 Additional issues created

- Issue #15 — Include required student fields and duplicate-ID rejection: supports F-001 and F-004.
- Issue #16 — Implement accessibility support: supports NF-005.
- Issue #17 — Implement performance requirements and response-time testing: supports NF-001 and NF-006.
- Issue #18 — Implement secure logout and session termination: supports F-024.

5.2 F-025: user-account management versus RBAC

RBAC (Issue #6) defines what each role is allowed to do. F-025 separately requires administrators to create, modify, and deactivate authorized user accounts. A clarification was requested on Issue #6 so these account-management actions have explicit acceptance criteria and verification tests. Until the issue and team confirm that scope, F-025 should be treated as partially covered rather than fully verified.

5.3 F-023: consistency of related academic records

Issue #10 covers data persistence and storage, and Issue #12 covers input validation and error handling. However, F-023 needs explicit rules for what happens to related grades and attendance when a student or subject is updated or deleted. A clarification was requested on Issue #10 to cover consistency, orphaned records, and corresponding test cases. This remains a verification item until the rules and test results are confirmed.

5.4 NF-002: reliability and recovery

Issue #10 is relevant to persistent storage, but persistence alone does not demonstrate reliability. NF-002 should remain partially covered until failure scenarios, unexpected termination, data-loss prevention, and recovery expectations are defined and tested with recorded evidence.

5.5 Issue #14: testing backlog completeness

Issue #14 is titled “Test functional and non-functional requirements.” Its scope should be reviewed to ensure it identifies the relevant requirements, test conditions, expected results, evidence to capture, and pass/fail recording. The existence of the issue or the test-case IDs in the RTM does not demonstrate that testing has been performed.

5.6 Performance and accessibility

Issue #17 links to the response-time and expected-volume requirements. The final verification should define the test dataset and load conditions, measure response times, and preserve results. Issue #16 links to accessibility; verification should include the applicable keyboard, focus visibility, screen-reader, contrast, and accessible-error criteria. Neither issue alone proves compliance.

6. Verification Status and Evidence

The RTM supplied for review assigns status N (Not Tested) to all requirement rows. This report preserves that status. A requirement should be marked Passed only after the associated test is executed against the agreed acceptance criteria and the result is recorded.

Evidence item	What should be recorded	Current reporting rule
Functional tests	Test case ID, preconditions, steps, expected result, actual result, pass/fail, date and evidence reference.	Do not claim a pass without an actual result.
Security tests	Authentication, role restrictions, session timeout, failed-login handling and audit behavior.	Issue descriptions are plans, not proof of implementation.
Performance tests	Dataset size, load conditions, number of requests, measured response times and percentile calculation.	Do not claim NF-001/NF-006 compliance without measurements.
Reliability/data integrity	Restart/failure scenarios, recovery behavior, record consistency and checks for orphaned records.	Keep NF-002 and F-023 partial until criteria and results are confirmed.
Accessibility checks	Keyboard navigation, focus visibility, contrast, screen-reader behavior and accessible error handling.	Do not claim accessibility compliance without a documented audit.

Traceability Gaps and Clarifications

1. Student Record Deletion (Issue #4)

The current list of functional requirements does not explicitly define student record deletion. Issue #4 therefore requires clarification against the approved SRS before full traceability can be established. The deletion operation should also be checked for its effect on related subject, grade and attendance records.

2. User Account Management (F-025)

F-025 specifies that administrators can create, modify and deactivate permitted user accounts. This requirement is distinct from role-based access control. Issue #6 primarily addresses role permissions; implementation and testing of the account-management lifecycle should be tracked separately or explicitly included in an appropriate issue.

3. Data Consistency and Reliability (F-023 and NF-002)

F-023 requires invalid references between student, subject, grade and attendance records to be rejected. Additional testing should verify that record updates and deletions do not leave inconsistent or orphaned related records. NF-002 also requires verification of reliable operation without unexpected termination or data loss under normal usage. These requirements should remain unverified until appropriate tests and evidence are recorded.

8. Conclusion

The Requirements Traceability Matrix maps the Student Record System's functional, non-functional and security requirements to the associated GitHub issues and specified test cases. The review identifies areas requiring clarification, including student record deletion, user account lifecycle management, data consistency and reliability verification.

All test cases remain Not Tested unless supported by recorded execution results and evidence. The presence of a GitHub issue indicates that work is tracked; it does not independently demonstrate implementation or successful verification. The RTM should be updated as development and testing progress so that requirement coverage, outstanding actions and verification results remain accurate.

9. References

- **Software Requirements Specification (SRS):** Project SRS document supplied for the Student Record System.
- **GitHub repository:** https://github.com/Student-Record-System-project/Student-Record-System
- **GitHub Issues:** https://github.com/Student-Record-System-project/Student-Record-System/issues
- **GitHub Project Board:** https://github.com/orgs/Student-Record-System-project/projects/2/views/1

Appendix A. RTM Status Definitions

Status	Meaning	When to use
N — Not Tested	No recorded execution result is available.	Use before the test is run or when evidence is missing.
P — Passed	The test was executed and the actual result met the expected result.	Use only with a recorded pass result and supporting evidence.
A — Action Required	A gap, failure, ambiguity or corrective action remains.	Use when follow-up is required; include the reason/action in comments.

Coverage assessment is separate from test status. A requirement can be mapped to a GitHub issue while still being partially covered or not yet tested.